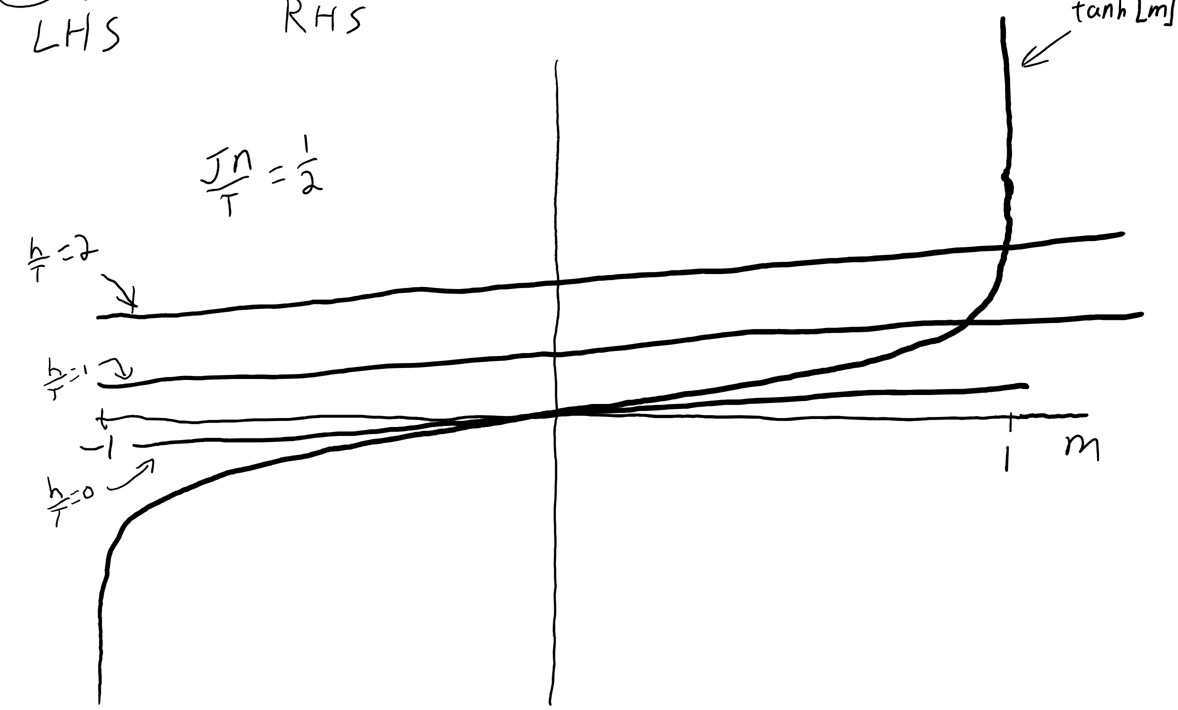
$\boxed{3.6.7}$

$$h = T \tanh^{-1}[m] - Jnm$$

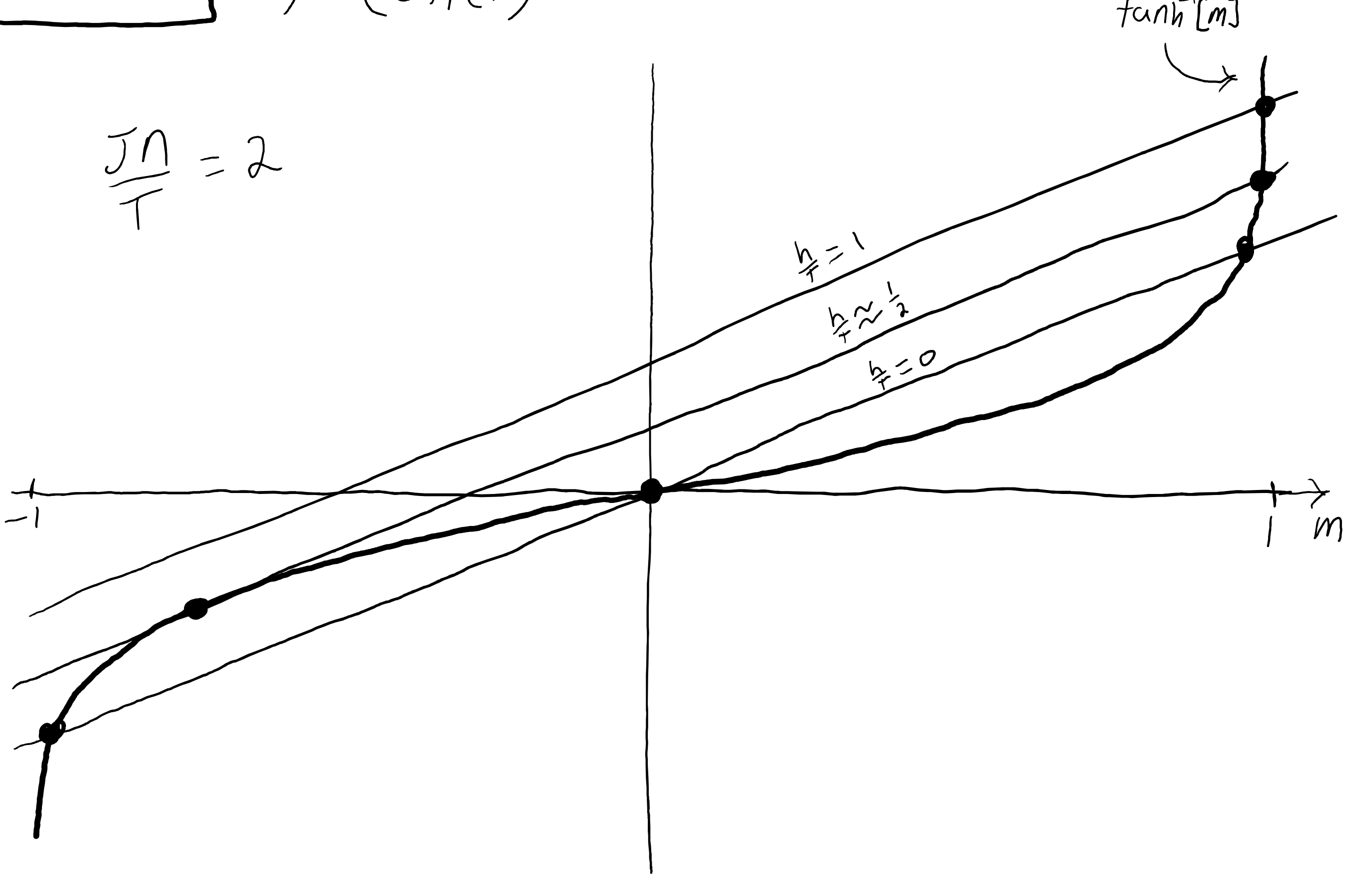
a) $$\underbrace{\frac{Jn}{T} m + \frac{h}{T}}_{LHS} = \underbrace{\tanh^{-1}[m]}_{RHS}$$



This is the qualitative picture for all $0 < \frac{Jn}{T} \leq 1$.

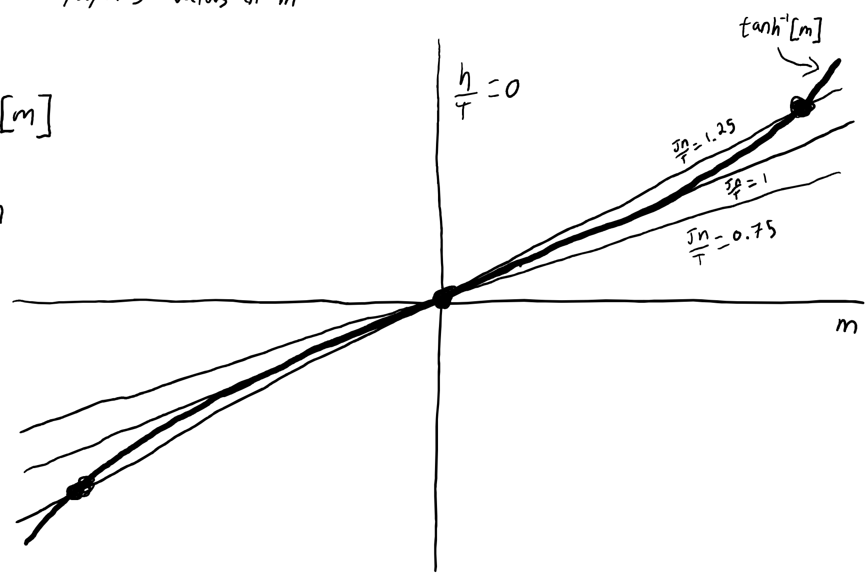
Note that there is only one value of m where LHS = RHS

$\boxed{3.6.7}$ a) (cont.)



This is the qualitative picture for $\frac{Jn}{T} > 1$.
Depending on the value of $\frac{h}{T}$, there can be 1, 2, or 3 values of m that satisfy the equation LHS = RHS.

b) $h = 0 \Rightarrow \frac{Jn}{T} m = \tanh^{-1}[m]$



The bifurcation will happen when $\frac{Jn}{T} m$ is tangential to $\tanh^{-1}[m]$.

$$\frac{d}{dm} \tanh^{-1}[m] \Big|_{x=0} = \frac{1}{1-x^2}\Big|_{x=0} = 1$$

so, when $\frac{Jn}{T} = 1$

or, $T_c = Jn$

3.7.6

$$\dot{x} = -kxy \qquad \dot{y} = kxy - ly \qquad \dot{z} = ly$$

a)

N is constant if $\dot{N} = 0$

$$N = x + y + z$$
$$\dot{N} = \dot{x} + \dot{y} + \dot{z} = -kxy + kxy - ly + ly = 0$$

b)

$$x(t) = x_0 e^{\frac{-kz(t)}{l}}$$

Assuming there are zero dead people at the start, $z(0) = 0$

$$x(0) = x_0 e^{\frac{-kz(t)}{l}} = x_0 e^0 = x_0$$
$$\dot{x}(t) = \frac{-k\dot{z}(t)}{l} x_0 e^{\frac{-kz(t)}{l}} = \frac{-kly}{l} x_0 e^{\frac{-kz(t)}{l}} = -kx_0 e^{\frac{-kz(t)}{l}} y = -kxy$$

c)

$$l\left(N - z - x_0 e^{\frac{-kz}{l}}\right) = l\left(x + y + z - z - x\right) = ly = \dot{z}$$

d)

$$\frac{\mathrm{d}z}{\mathrm{d}t} = l\left(N - z - x_0 e^{\frac{-kz}{l}}\right)$$
$$\frac{1}{kx_0}\frac{\mathrm{d}\frac{kz}{l}}{\mathrm{d}t} = \frac{N}{x_0} - \frac{l}{kx_0}\frac{kz}{l} - e^{\frac{-kz}{l}}$$
$$u = \frac{kz}{l} \qquad \tau = kx_0 t \qquad a = \frac{N}{x_0} \qquad b = \frac{l}{kx_0}$$
$$\frac{\mathrm{d}u}{\mathrm{d}\tau} = a - bu - e^{-u}$$

e)

$a = \frac{N}{x_0}$ which is the total number of people divided by the number of healthy people. There can't be more healthy people than there are people, so $x_0 \leq N$ and $1 \leq \frac{N}{x_0} = a$

$0 < l,\ k,\ x_0 \Rightarrow 0 < \frac{l}{kx_0} = b$

f)

$$\frac{\mathrm{d}}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}\tau} = \frac{\mathrm{d}}{\mathrm{d}u}\left(a - bu - e^{-u}\right) = -b + e^{-u} = 0 \Rightarrow u = \ln\left(\frac{1}{b}\right)$$
$$\frac{\mathrm{d}^2}{\mathrm{d}u^2}\frac{\mathrm{d}u}{\mathrm{d}\tau} = -e^{-u} < 0$$

The is only one critical point of $\frac{\mathrm{d}u}{\mathrm{d}\tau}$ which is a local max, and the second derivative is always negative, so there should be two fixed points to the left and right of $u = \ln\left(\frac{1}{b}\right)$ which are unstable and stable respectively.

3.7.6 cont.

g)

The maximum of $\dot{u}$ will occur at an inflection point

$$\begin{aligned}
u(\tau) &= u(kx_0t) = \frac{kz(t)}{l} \\
\frac{\mathrm{d}}{\mathrm{d}t}u(\tau) &= kx_0\frac{\mathrm{d}u}{\mathrm{d}\tau}(kx_0t) = \frac{k\dot{z}(t)}{l} = ly \\
\frac{\mathrm{d}^2}{\mathrm{d}t^2}u(\tau) &= (kx_0)^2\frac{\mathrm{d}^2u}{\mathrm{d}\tau^2}(kx_0t) = \frac{k\ddot{z}(t)}{l} = l\dot{y} = 0
\end{aligned}$$

So the maximum of $\dot{u}$, $\dot{z}$, and y all occur at the same time. This time t_{peak} is shown to exist by finding the maximum of $\frac{\mathrm{d}u}{\mathrm{d}\tau}$ in terms of u, which occurs at an inflection point.

$$\frac{\mathrm{d}^2u}{\mathrm{d}\tau^2} = \frac{\mathrm{d}}{\mathrm{d}\tau}\left(a - bu - e^{-u}\right) = -b\frac{\mathrm{d}u}{\mathrm{d}\tau} + \frac{\mathrm{d}u}{\mathrm{d}\tau}e^{-u} = \frac{\mathrm{d}u}{\mathrm{d}\tau}\left(-b + e^{-u}\right)$$

So $\frac{\mathrm{d}u}{\mathrm{d}\tau}$ has inflection points at $u = \ln\left(\frac{1}{b}\right)$ and the roots of the $\frac{\mathrm{d}u}{\mathrm{d}\tau}$ equation. The roots are defined as $\frac{\mathrm{d}u}{\mathrm{d}\tau} = 0$ so the maximum occurs at $u = \ln\left(\frac{1}{b}\right)$

h)

Assuming there are zero dead people at the start, $u(0) = 0$

$$\left.\frac{\mathrm{d}u}{\mathrm{d}\tau}\right|_{t=0} = a - bu(0) - e^{-u(0)} = a - b(0) - e^0 = a - 1 > 0$$

$\dot{u}$ reaches a maximum at $t = t_{\text{peak}}$ when $u = \ln\left(\frac{1}{b}\right) > 0$ if $b < 1 \Rightarrow 0 < t_{\text{peak}}$ since $u(0) = 0$ and u is increasing.

$\dot{u} \to 0$ because u approaches a stable fixed point, meaning eventually people stop dying.

i)

$\dot{u}$ reaches a maximum at $t = t_{\text{peak}}$ when $u = \ln\left(\frac{1}{b}\right) < 0$ if $1 < b \Rightarrow t_{\text{peak}} < 0$ since $u(0) = 0$ and u is increasing. However, a negative time for t_{peak} doesn't fit the model where $0 < t$ so we take take the initial time $t_{\text{peak}} = 0$ as the maximum for $\dot{u}$.

j)

$b = \frac{l}{kx_0}$ is a ratio of the sick rate and the death rate. If people die soon after getting sick, then the disease will have trouble spreading. However, the relationship is a little more complicated because the sick rate is proportional to the population size, so a large population will be capable of an epidemic more than the same disease in a small population. $b = 1$ is the critical value at which the spread of the disease will be speeding up or slowing down for $b < 1$ or $1 < b$ respectively.

k)

This model ignores other slower factors in population growth such as birth rate and unrelated death rate, and a model for AIDS would need to take into account the much slower time scale of the disease. This could be done by adding another term to the $\dot{x}$ and $\dot{z}$ equations which is proportional to the total population.

$\boxed{4.1.8}$ a) $\dot{\theta} = \cos[\theta] = -\frac{dV}{d\theta}$

$$\Rightarrow V[\theta] = -\sin[\theta] + c$$

$$V[\theta + 2\pi k] = -\sin[\theta + 2\pi k] + c = V[\theta]$$

b) $\dot{\theta} = 1 = -\frac{dV}{d\theta} \Rightarrow V[\theta] = -\theta + c$

$$V[\theta + 2\pi k] = -(\theta + 2\pi k) + c = V[\theta] - 2\pi k \neq V[\theta]$$

c) In addition to $f(\theta)$ being 2π periodic and smooth (so we can garuntee existance and uniquness), we also need $V(2\pi) = V(0)$. This will be the case if

$$\int_0^{2\pi} d\theta\, f(\theta) = 0$$

Since

$$\int_0^{2\pi} d\theta\, f(\theta) = \int_0^{2\pi} d\theta \left(-\frac{dV}{d\theta}\right) = -V(2\pi) - V(0) = 0$$

$$\Rightarrow V(2\pi) = V(0).$$

$\boxed{4.2.2}$

$$2i(\sin[\theta]+\sin[\phi]) = (e^{i\theta}-e^{-i\theta}) + (e^{i\phi}-e^{-i\phi})$$

$$= e^{i\theta} - e^{-i\phi} + e^{i\phi} - e^{-i\theta}$$

$$= e^{i(\frac{\theta}{2}+\frac{\theta}{2}+\frac{\phi}{2}-\frac{\phi}{2})} - e^{i(\frac{\theta}{2}-\frac{\theta}{2}-\frac{\phi}{2}-\frac{\phi}{2})} + e^{i(\frac{\theta}{2}-\frac{\theta}{2}+\frac{\phi}{2}+\frac{\phi}{2})} - e^{-i(\frac{\theta}{2}+\frac{\theta}{2}+\frac{\phi}{2}-\frac{\phi}{2})}$$

$$= e^{i(\frac{\theta+\phi}{2})}e^{i(\frac{\theta-\phi}{2})} - e^{-i(\frac{\theta+\phi}{2})}e^{i(\frac{\theta-\phi}{2})} + e^{i(\frac{\theta+\phi}{2})}e^{-i(\frac{\theta-\phi}{2})} - e^{-i(\frac{\theta+\phi}{2})}e^{-i(\frac{\theta-\phi}{2})}$$

$$= \left(e^{i(\frac{\theta+\phi}{2})} - e^{-i(\frac{\theta+\phi}{2})}\right)\left(e^{i(\frac{\theta-\phi}{2})} + e^{-i(\frac{\theta-\phi}{2})}\right)$$

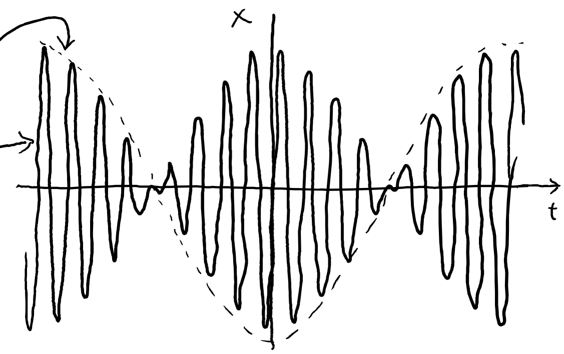
$$= 2i\sin\left[\frac{\theta+\phi}{2}\right] 2\cos\left[\frac{\theta-\phi}{2}\right]$$

$$\Rightarrow \sin[\theta]+\sin[\phi] = 2\sin\left[\frac{\theta+\phi}{2}\right]\cos\left[\frac{\theta-\phi}{2}\right]$$

a)

$$\sin[8t]+\sin[9t] = 2\sin\left[\frac{8+9}{2}t\right]\cos\left[\frac{9-8}{2}t\right]$$

$$= 2\underbrace{\sin\left[\frac{17}{2}t\right]}_{\text{oscillation}}\underbrace{\cos\left[\frac{1}{2}t\right]}_{\text{modulation}}$$



$\Rightarrow A = 2$

$T = \frac{2\pi}{\omega_m} = \frac{2\pi}{\frac{1}{2}} = 4\pi$

(The beat is always twice the freq of the modulation)

b) See the math above part (a).

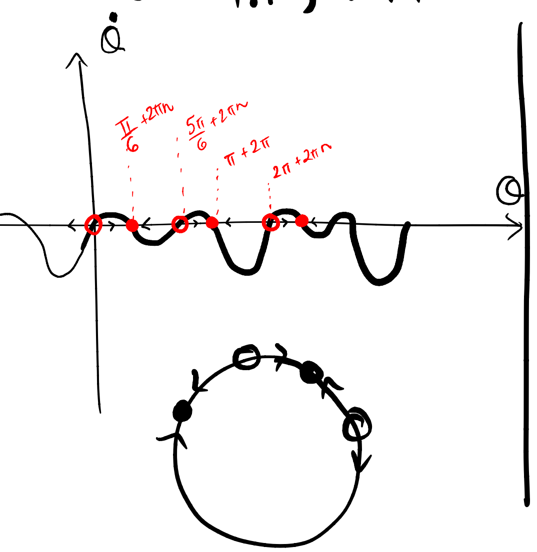
4.3.6 $\dot{\theta} = \mu + \sin\theta + \cos 2\theta$

$\mu = 0$,

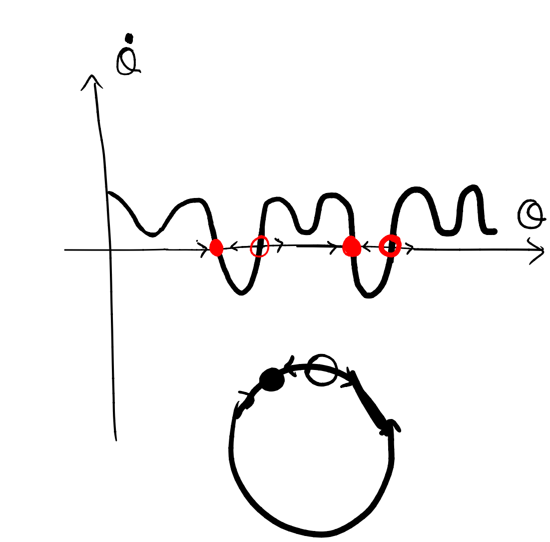
$\theta^* = \frac{7\pi}{6} + 2\pi n$, stable

$\theta^* = \frac{11\pi}{6} + 2\pi n$, unstable

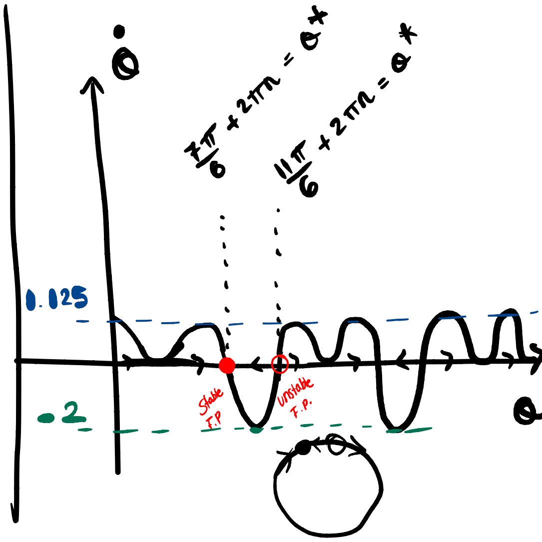
$\mu = -1.1$; 4 FP



$\mu = 1$; 2 FP



$\mu = 0$; 2 FP



$$\boxed{\text{Bifurcation } \mu: \ -1.125 < \mu < 2 \text{ ; outside of this range, no FP occur.}}$$

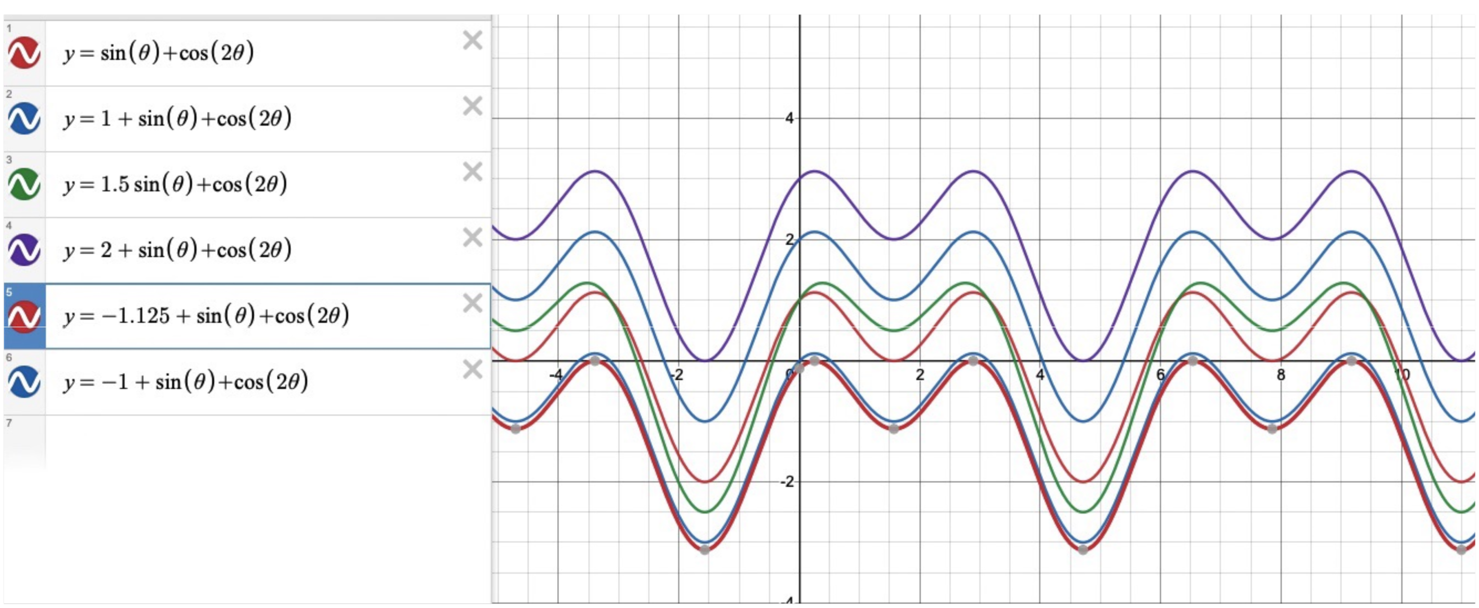
$\ddot{\theta} = \cos\theta - 2\sin 2\theta$

(obtained from graphical analysis ; plotting & observing)

$$\boxed{f(\theta) \text{ @ critical points} = 1.125 \text{ \& } -2 \text{ \& } 0}$$

$$\boxed{\text{Saddle Node Bifurcations occur @ those } \mu \text{ values}}$$

Bifurcations with $-1.125 < \mu < 2$:



$\boxed{4.4.1}$

$$mL^2\ddot{\theta} + b\dot{\theta} + mgL\sin[\theta] = \Gamma$$

Start by rewrighting this diff. eq. to have a dimensionles paramater on the term we want to remove. To do this, we can rescale time by

$$\tau = tT$$

$$T^2 mL^2 \frac{d^2}{d\tau^2}\theta + Tb\frac{d}{d\tau}\theta + mgL\sin[\theta] = \Gamma$$

$$T^2 \frac{L}{g}\frac{d^2\theta}{d\tau^2} + T\frac{b}{mgL}\frac{d\theta}{d\tau} + \sin[\theta] = \frac{\Gamma}{mgL}$$

now, choose the time scale, T, to be

$$T = \frac{mgL}{b}$$

$$\Rightarrow \frac{gm^2L^3}{b^2}\frac{d^2\theta}{d\tau^2} + \frac{d\theta}{d\tau} + \sin[\theta] = \frac{\Gamma}{mgL}$$

so, if $\frac{gm^2L^3}{b^2} >> 1$ this reduces to

$$\frac{d\theta}{d\tau} + \sin[\theta] = \frac{\Gamma}{mgL}$$

so,

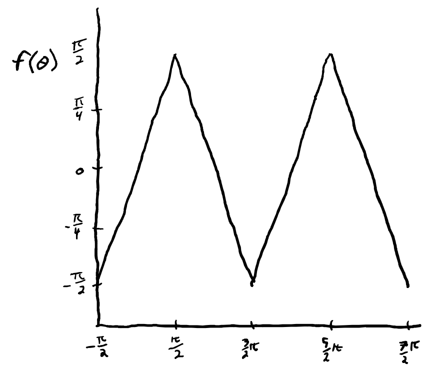
$$\frac{b}{mgL}\frac{d\theta}{dt} + \sin[\theta] = \frac{\Gamma}{mgL}$$

or,

$$\boxed{b\dot{\theta} + mgL\sin[\theta] = \Gamma}$$

$\boxed{4.5.1}$

a)



b) In the range of entrainment, a firefly will synchronize. This means $\dot{\phi} = \dot{\Theta} - \dot{\theta} = 0$

So, $\Omega - \omega - Af(\Theta - \theta) = 0 \Rightarrow \Omega = \omega + Af(\Theta - \theta)$

and since $f(\theta)$ ranges from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$

$$\omega - A\frac{\pi}{2} \leq \Omega \leq \omega + A\frac{\pi}{2}$$

c) Phase-locked means $\dot{\phi} = \Omega - \omega - Af(\phi^*) = 0 \Rightarrow f(\phi^*) = \frac{\Omega - \omega}{A}$

d) $$T_{drift} = \int_0^{2\pi} dt = \int_0^{2\pi} d\phi \frac{dt}{d\phi} = \int_0^{2\pi} d\phi \frac{1}{\dot{\phi}} = \int_0^{2\pi} d\phi \frac{1}{\Omega - \omega - Af(\phi)}$$

$$= \frac{1}{A}\int_{-\frac{\pi}{2}}^{\pi/2} d\phi \frac{1}{\frac{\Omega-\omega}{A} - \phi} + \frac{1}{A}\int_{\frac{\pi}{2}}^{\frac{3}{2}\pi} d\phi \frac{1}{\frac{\Omega-\omega}{A} - \pi + \phi}$$

$$= \frac{2}{A}\ln\left[\frac{\frac{\Omega-\omega}{A} + \frac{\pi}{2}}{\frac{\Omega-\omega}{A} - \frac{\pi}{2}}\right]$$